



Kali Linux Target VM Target VM

10.0.0.2 10.0.0.3 10.0.0.4

Attacker Web App Linux VM

■■ Lab Configuration

Note: Network addresses, gateway addresses, DNS servers, and interface names may vary depending on the VirtualBox and Kali Linux configuration.

■ Tools Used

■ Setup Procedure

1. Install 7-Zip

7-Zip was installed to extract the Kali Linux virtual-machine package distributed as a compressed archive.

Purpose:

Extract Kali Linux VM files.

Prepare the VM for import into VirtualBox.

2. Install VirtualBox

VirtualBox was installed as the virtualization platform used to host the cybersecurity laboratory.

The virtualization environment provides the ability to:

Create virtual machines.

Configure virtual networking.

Allocate system resources.

Create snapshots.

Restore clean VM states.

3. Create the NAT Network

A dedicated NAT Network named:

NatNetwork

was created in VirtualBox.

Network Configuration

Network Name: NatNetwork

IPv4 Prefix: 10.0.0.0/24

DHCP: Enabled

IPv6: Disabled

A NAT Network was selected because multiple virtual machines connected to the same NAT Network can communicate with one another while also having outbound network connectivity.

This configuration provides a practical foundation for building a multi-machine cybersecurity laboratory.

4. Import Kali Linux

The Kali Linux virtual machine was imported into VirtualBox.

VM Configuration

Operating System: Kali Linux 2026.2

RAM: 2048 MB

Network Adapter: Adapter 1

Attached To: NAT Network

Network: NatNetwork

Adapter Type: Intel PRO/1000 MT Desktop

A shared folder was also configured to facilitate controlled file transfer between the host operating system and the Kali Linux VM.

5. Configure Kali Linux Networking

The Kali Linux network configuration was checked and configured with a consistent IPv4 address.

Example Configuration

IP Address: 10.0.0.2

Subnet Mask: 255.255.255.0

Gateway: 10.0.0.1

DNS: 8.8.8.8

A consistent address makes it easier to reference the Kali machine during future laboratory exercises.

Verify the IP Address

```
ip a
```

Check the Routing Table

```
ip route
```

Check DNS Configuration

```
resolvectl status
```

■ Lab Verification

After configuring the environment, several tests were performed to verify the laboratory.

Example Network Result

IP Address: 10.0.0.2/24

Gateway: 10.0.0.1

DNS: 8.8.8.8

■ Verification Commands

Check Network Interfaces

```
ip a
```

Check Routing

```
ip route
```

Test the Virtual Gateway

```
ping 10.0.0.1
```

Test External Connectivity

```
ping 8.8.8.8
```

Test DNS Resolution

```
nslookup networkwalks.com
```

Verify Nmap

```
nmap --version
```

These commands provide basic verification that the virtual laboratory is functioning correctly.

■ Troubleshooting

Problem 1 — Internet Connectivity After Static IP Configuration

After manually configuring IPv4 settings, Internet connectivity may fail depending on the Kali Linux and NetworkManager configuration.

One workaround used during the laboratory setup was:

```
sudo nmcli connection modify "Wired connection 1" ipv4.dad-timeout 0
```

The connection was then restarted or the system was rebooted before connectivity was tested again.

Important

Network connection names may differ between systems.

Before modifying a connection, identify the actual connection name:

```
nmcli connection show
```

Then apply the configuration to the correct connection.

This setting is a troubleshooting workaround, not a universal solution for network connectivity problems.

■ Problem 2 — VirtualBox VT-x / Hardware Virtualization Error

The Kali Linux VM initially failed to start because hardware virtualization was disabled in the system firmware.

Resolution

Restart the computer.

Enter BIOS/UEFI settings.

Locate the virtualization setting.

Enable Intel VT-x / hardware virtualization.

Save the BIOS/UEFI configuration.

Restart the computer.

Start the Kali Linux VM again.

After enabling hardware virtualization, the VM started successfully.

■ Screenshots

Screenshots documenting the laboratory should be stored in the screenshots/ directory.

Recommended evidence:

screenshots/

■■■■ virtualbox-nat-network.png

■■■■ kali-vm-settings.png

■■■■ kali-network-config.png

■■■■ ip-a.png

■■■■ ip-route.png

■■■■ ping-gateway.png

■■■■ internet-connectivity.png

■■■■ dns-resolution.png

■■■■ nmap-version.png

■■■ clean-snapshot.png

Screenshots provide evidence that the configuration and verification steps were actually performed.

■ Snapshot & Recovery

A clean VirtualBox snapshot was created after completing the initial laboratory configuration.

Snapshot Name

Clean Kali - Network Setup

The snapshot represents the known-good baseline of the laboratory.

Before performing potentially disruptive experiments, the VM can be restored to this snapshot.

Recovery Workflow

Clean Baseline



Create Snapshot



Perform Security Experiment



Configuration Changed?



YES NO



Restore Continue

Snapshot



Known-Good State

Snapshots are particularly useful when experimenting with security tools, configurations, vulnerable applications, and networking changes.

■ Key Concepts Learned

1. NAT vs NAT Network

A standard VirtualBox NAT configuration and a NAT Network are not identical.

A NAT Network allows multiple virtual machines connected to the same virtual network to communicate with one another while providing network address translation for outbound connectivity.

This makes NAT Networks useful when creating multi-machine cybersecurity laboratories.

2. Virtual Machine Networking

This project provided practical experience with:

Virtual network adapters

NAT Networks

IPv4 addressing

Subnet masks

Default gateways

DNS

Network connectivity testing

Virtual machine communication

3. Static IPv4 Configuration

I learned how IPv4 configuration affects network communication and how to verify:

IP Address

Subnet Mask

Default Gateway

DNS

Routing

I also learned that manually configured network settings must be consistent with the virtual network's addressing scheme.

4. VM Snapshots

I learned the importance of creating a clean snapshot before performing risky or experimental activities.

A known-good snapshot provides a reliable recovery point and makes repeated cybersecurity experiments easier to manage.

5. Troubleshooting

The project provided practical troubleshooting experience involving:

Hardware virtualization

BIOS/UEFI settings

VirtualBox networking

Kali Linux networking

NetworkManager

IPv4 configuration

Connectivity testing

6. Technical Documentation

I learned that professional cybersecurity work involves more than executing commands.

Good documentation should capture:

Configuration

Commands

Results

Screenshots

Problems

Solutions

Lessons learned

Security considerations

This makes technical work reproducible and easier for others to understand.

■ Security & Ethical Use

This laboratory is intended strictly for education, authorized security testing, and cybersecurity research.

All scanning, exploitation, vulnerability assessment, and penetration-testing activities must be performed only against:

Systems you own.

Systems specifically created for testing.

Systems for which you have explicit authorization.

Do not use the tools or techniques demonstrated in this laboratory against unauthorized systems, networks, websites, or devices.

■ Future Improvements

The laboratory will be expanded as additional cybersecurity projects are completed.

Planned improvements include:

Add an intentionally vulnerable Linux target VM.

Add a vulnerable Windows target VM.

Deploy a vulnerable web application.

Practice network reconnaissance.

Perform controlled Nmap scans.

Perform vulnerability assessments.

Analyze network traffic with Wireshark.

Practice web security testing.

Document controlled exploitation scenarios.

Add network diagrams.

Automate selected lab configuration tasks.

Create individual reports for future security assessments.

■ Resources

7-Zip

VirtualBox

Kali Linux

Nmap

■ Repository Structure

cybersecurity-lab/

■

■■■■ README.md

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■■■■ documentation/

■ ■■■■ lab-architecture.md

- ■■■■ setup-procedure.md
- ■■■■ troubleshooting.md
- ■■■■ verification.md
-
- screenshots/
- ■■■■ virtualbox-nat-network.png
- ■■■■ kali-vm-settings.png
- ■■■■ kali-network-config.png
- ■■■■ ip-a.png
- ■■■■ ip-route.png
- ■■■■ ping-gateway.png
- ■■■■ dns-resolution.png
- ■■■■ nmap-version.png
- ■■■■ clean-snapshot.png
-
- diagrams/
- lab-architecture.png
- Project Information

Program: Cybersecurity at Networkwalks

Week: 01

Project: Cybersecurity & Penetration Testing Lab Setup

Focus: Virtualization, Linux Networking & Cybersecurity Lab Architecture

Platform: VirtualBox + Kali Linux

Repository Owner: yarimahafiz001-ops

■■■■ Author

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Interested in:

Cybersecurity

Ethical Hacking

Network Security

Vulnerability Assessment

Information Technology

Security Research

Continuous Technical Learning

■ Project Status

Status: ■ Completed — Initial Lab Setup

The laboratory is ready for future authorized cybersecurity exercises and additional target-machine deployments.

Component	Configuration
Host OS	Windows 10
Processor	Intel Core i7
Host RAM	8 GB
Hypervisor	VirtualBox 7.2
Security OS	Kali Linux 2026.2
Kali RAM	2048 MB
Virtual Network	NAT Network
Network Name	NatNetwork
Network Address	10.0.0.0/24
Kali IPv4 Address	10.0.0.2/24
Default Gateway	10.0.0.1
DNS Server	8.8.8.8
DHCP	Enabled
IPv6	Disabled
Future VM Range	10.0.0.3–10.0.0.99

Tool	Purpose
7-Zip	Extracting compressed Kali VM files
VirtualBox	Virtualization and VM management
Kali Linux	Cybersecurity testing platform
Nmap	Network discovery and port scanning
NetworkManager / nmcli	Linux network configuration
ping	Network connectivity testing
ip	Network interface and routing inspection
nslookup	DNS resolution testing

Test	Command	Expected Result
Check IP address	ip a	Kali IP is displayed
Check routing	ip route	Correct gateway/route displayed
Test gateway	ping 10.0.0.1	Successful replies
Test Internet	ping 8.8.8.8	Successful replies
Test DNS	nslookup networkwalks.com	Domain resolves
Verify Nmap	nmap --version	Nmap version displayed
Verify snapshot	Restore snapshot + ip a	Baseline configuration restored